Ahmed Ghaben

Applied Mathematics & Computer Science · MIPT

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PROFILE

Third-year Applied Mathematics and Computer Science student at MIPT. My research focuses on diverse solutions to combinatorial optimization problems, and I build full-stack software in Python, C++ and TypeScript. Seeking an internship in quantitative research or software engineering.

EDUCATION

Moscow Institute of Physics and Technology (MIPT) 2024 – 2028 (expected)

BSc Applied Mathematics and Computer Science, English-taught program · GPA 9.14 / 10 (3.92 / 4.0)

RESEARCH

Supervised by Danil Sagunov. Started at LIPS 2026, MIPT's summer research program, and presented at its conference.

Diverse matroid bases and spanning trees

Manuscript in preparation · 2026

- Proved that finding three matroid bases with maximum minimum pairwise distance is solvable in polynomial time with O(r) oracle calls, and in O(n) time for spanning trees.
- Proved the same for four diverse spanning trees and extended the four-basis result to general matroids.

Diverse maximum cuts in planar graphs

Manuscript in preparation · 2026

- Characterized when a graph has three maximum cuts at the largest possible pairwise distance, reducing a search over triples to a search over pairs.
- For planar graphs, showed this holds exactly when the bound $3 \cdot OPT \geq 2m$ is tight, making this case of an open problem decidable in polynomial time; verified computationally on more than 23,000 graphs.

PROJECTS

Cloud point-of-sale system GitHub

Founder · in development

- Multi-store POS platform covering checkout, inventory, purchasing, returns, staff and reporting (TypeScript, React, NestJS, PostgreSQL, Prisma).
- AI store briefing (Claude API) that summarizes performance and suggests what to restock.
- Role-based access with per-user permissions, refresh-token rotation with reuse detection, audit log, unit tests, CI and Docker builds.

MyProfit: finance assistant and trading bot GitHub

Co-developer · 2025

- Built the financial engine (expense analysis, monthly reports, savings and trading-budget recommendations), the Telegram bot interface, the integration with the trading engine, and trade reporting (Python).

Bura card game GitHub

Solo · in development

- Python and Pygame game with tested logic separate from the interface; computer and LLM opponents planned.

EXPERIENCE

Mathematics and ICT Teacher (part-time) · Riverside International School

Sep 2025 – present

- Teach IGCSE Mathematics and ICT to Year 9 and 10 students alongside full-time study.

LEADERSHIP AND ACTIVITIES

- **Organizer, Phystech Buddy (MIPT):** help new students settle in and guide them through their first year; organize events for incoming students.
- **Sports:** player on the MIPT football team; certified swimming lifeguard.

SKILLS

Mathematics	Probability, linear algebra, optimization, algorithms, graph theory
Programming	Python, C++, TypeScript
Tools	React, NestJS, PostgreSQL, Docker, Git, pytest
Languages	Arabic (native), English (fluent), Russian (intermediate)